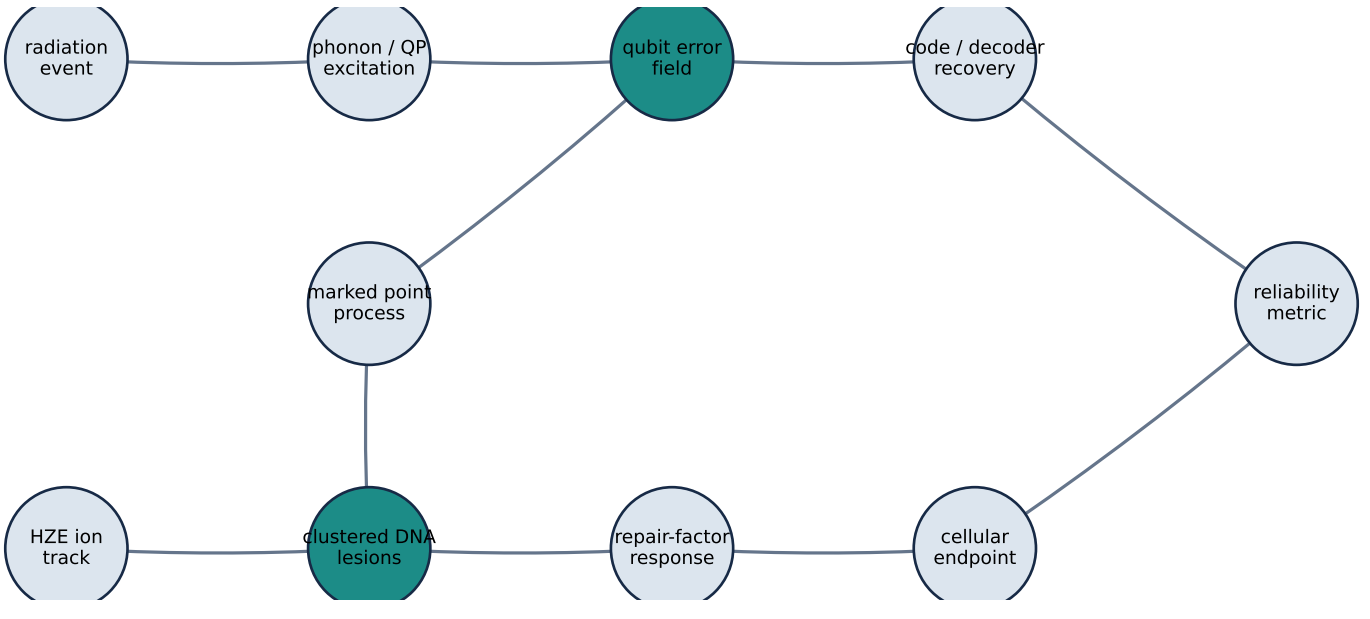
Domain-specific causal pathways and a shared statistical layer
radiation event
phonon / QP excitation
qubit error field
code / decoder recovery
marked point process
reliability metric
HZE ion track
clustered DNA lesions
repair-factor response
cellular endpoint